

(Mostly) OOP Problem Set

Throughout the textbook object-oriented programming (OOP) techniques have been used, but we mostly skipped those during the course. These questions are about some of the major ideas for using OOP to implement data structures and algorithms.

Questions asked on exams could be of any type, and will be variations of questions below, or questions that are similar.

Important: *Treat these problem-sets as non-AI activities!* Turn off all AI support and try to figure them out yourself **by hand**. Having AI or another student do this for you will not help you learn. You must do the learning yourself!

Question 1

This is not about OOP, but it is a topic covered in the lecture.

- Write the recurrence relation for the Fibonacci numbers, such that $F_1 = 1$, $F_2 = 1$, and so on.
- Prove that $F_n \leq 2^n$, where F_n is the n th Fibonacci number.
- You can calculate the numbers of the Fibonacci sequence using the recurrence relation, but that results in an exponential number of function calls. Explain what **memorization** is, and how it can be used to speed up calculating the Fibonacci sequence.

Question 2

Using chapter 2 of the textbook as a guide, explain the following OOP concepts: **abstraction**, **encapsulation**, and **modularity**.

Question 3

Explain and give code examples of the following C++ concepts:

- **virtual methods**
- **virtual destructors**
- **pure methods**

Question 4

Give a C++ **abstract base class** (including a virtual destructor, but with no other implementation details) for the essential operations of a **queue** of strings.

Question 5

- a) What is an **iterator**?
- b) Give an abstract base class of a for an iterator class that works with a collection of items of type `T` and allows you to get the next item from the iterator, and can also tell you if the iterator has more values.
- c) Write a function that uses an iterator (as defined in the previous question) to return `true` if the given value `x` (of type `T`) is contained in the collection the iterator refers to, and `false` otherwise.